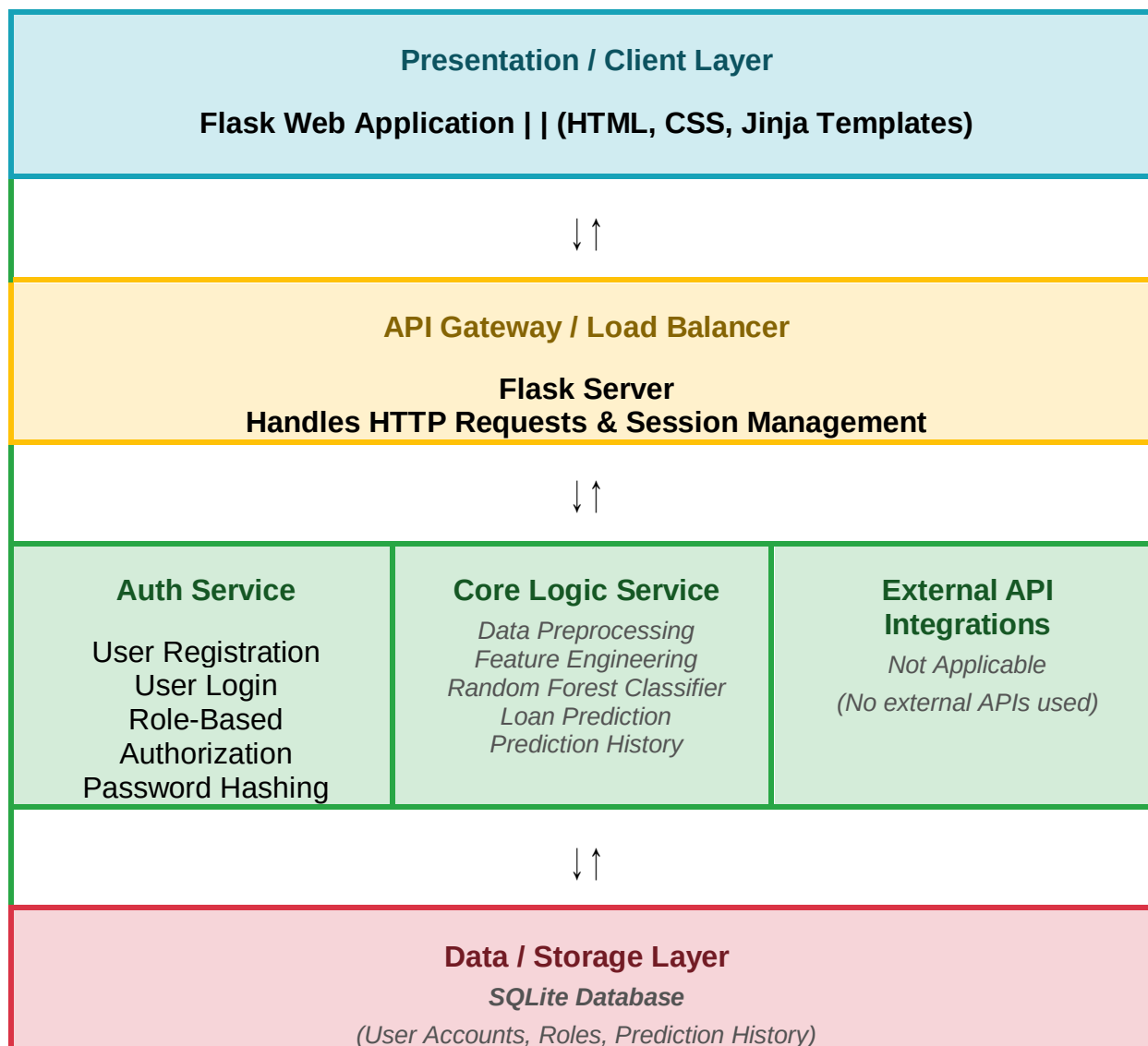


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Smart Lender – Loan Eligibility Prediction System follows a layered architecture consisting of a Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a user-friendly web interface for entering applicant details. The Flask application processes user requests and communicates with the machine learning model. The Core Logic Layer performs data preprocessing and prediction using a trained Random Forest classifier, while the Data Layer stores the loan dataset and trained model files required for generating loan eligibility predictions.



Loan Dataset (CSV)
Trained ML Model (.pkl)
Joblib Model Storage

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides a web-based interface where users enter applicant information and view the loan eligibility prediction results.</i>	<i>HTML5, CSS3, Jinja2 Templates</i>
<i>[API Gateway]</i>	<i>Handles HTTP routing, request processing, session management, authentication, authorization, and communication between application components.</i>	<i>Python, Flask</i>
<i>[Microservices]</i>	<i>Performs data preprocessing, feature engineering, executes the trained Random Forest model, generates loan eligibility predictions, and manages prediction history.</i>	<i>Scikit-learn, Pandas, NumPy, Joblib</i>
<i>[Database]</i>	<i>Stores the loan dataset used for training and the serialized trained machine learning model used during prediction.</i>	<i>SQLite, CSV Files, Joblib (.pkl)</i>